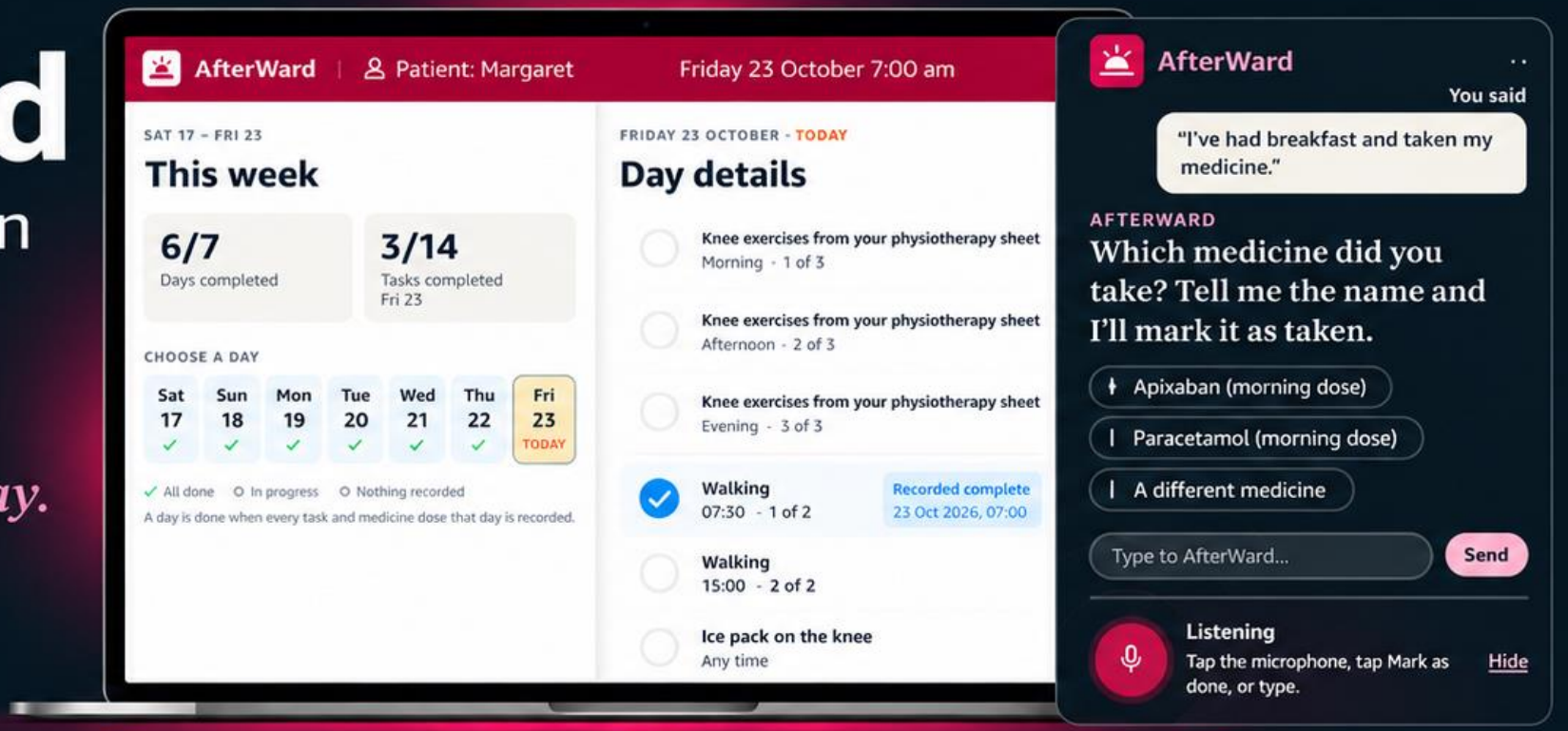
BUILD, SHIP, SHAPE: AMAZON DEVELOPER HACKATHON

AfterWard

A Living Recovery Plan for Alexa+

Discharge is one moment. Recovery happens every day.

AfterWard turns a discharge letter into a personalized, day-by-day plan that updates as you recover.



Alexa skill
Classic skill, and a simulated Alexa+ web app

MCP
9 tools for a living recovery plan

AWS
Secure, scalable cloud architecture

Open Source
Reusable validation package

THE NEED

What inspired us

The care team stays at the hospital. The paper goes home.

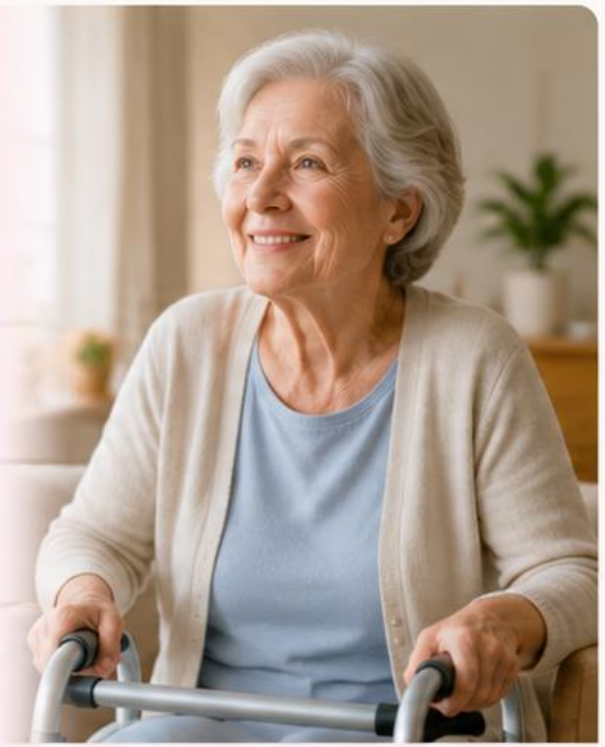
From that moment, the patient has to remember which medicine is due, what they are allowed to do, when the follow-up is and which warning signs matter. They do this while tired, sore or in pain.

The information is all on the page. But paper cannot notice that you already did your morning walk. We wanted to help with the part of recovery that happens at home — the everyday moments that matter, and make that support available to people of any age, anywhere in the world.

WHO WE BUILD FOR

Margaret, 68

- A new knee and a walking frame
- Managing recovery at home, on her own
- Her hands are on the frame, so talking is easier than tapping
- Wants to recover safely and stay independent



A synthetic patient. No real patient data is used anywhere.

THE CHALLENGE AT HOME

Remember medicines
Which medicine is due and for how long?

Know what's allowed
What activities are safe and what to avoid?

Keep follow-ups
When is the next appointment and why?

Spot warning signs
Which symptoms need urgent help?

THE NEED

Paper goes home. The care team stays behind.

A discharge letter holds every instruction, but it cannot support each moment of recovery at home.

Paper cannot remind her

The morning walk, the tablets after breakfast and the follow-up dates still depend on memory.

Paper cannot remember

It never changes when Margaret finishes a walk or takes a dose.

Paper cannot clarify

An unclear end date or two different dates needs a person, not a confident guess.

WHO WE BUILD FOR

Margaret, 68

- A new knee and a walking frame
- Managing recovery at home, on her own
- Her hands are on the frame, so talking is easier than tapping

A synthetic patient. No real patient data is used anywhere.



We wanted to help with the part of recovery that happens at home.

The everyday moments that matter, and make that support available to people of any age, anywhere in the world.

IN 2023 · IN THE US ALONE

nearly 3.8 million

adult hospital readmissions within 30 days

a 13.5% readmission rate for adult hospital stays

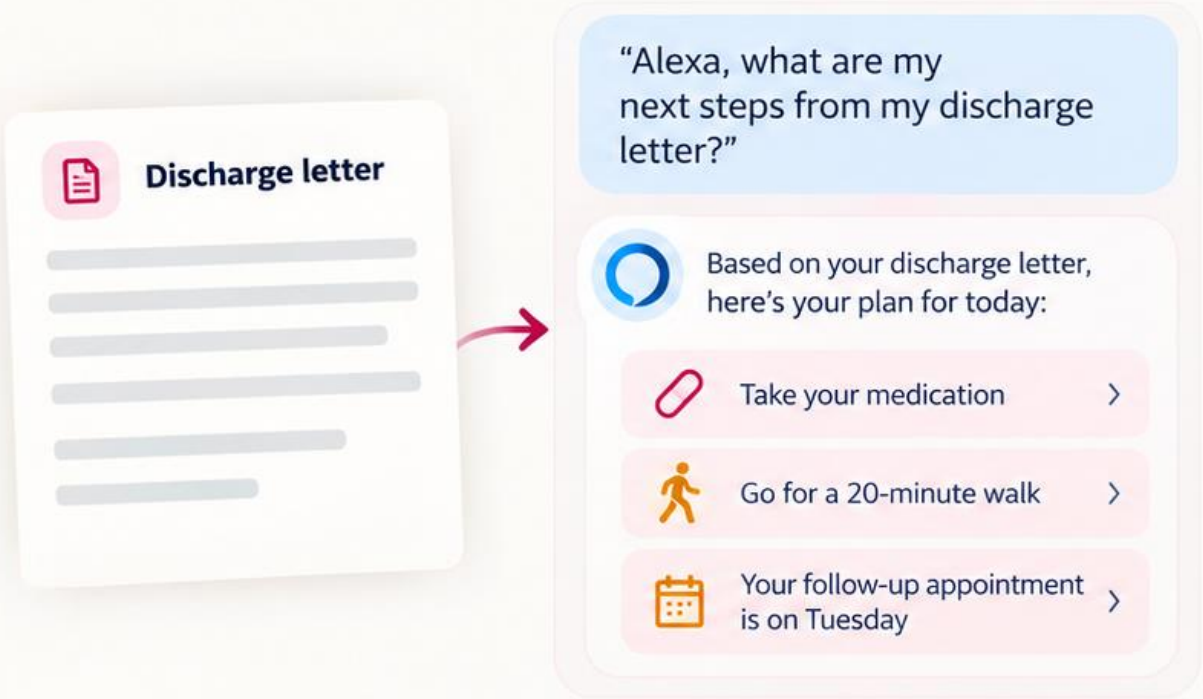
Source: AHRQ HCUP, Nationwide Readmissions Database, 2023 (Table A6): 3,769,100 readmissions

hcup-us.ahrq.gov/db/nation/nrd/NRDIntroduction2023.pdf

THE PRODUCT: WHY THIS IS A NEW EXPERIENCE

What AfterWard adds to Alexa+

From a simple reminder to a personalized, evolving recovery plan.



AFTERWARD ADDS

A PLAN BUILT FOR YOU, FROM YOUR DISCHARGE LETTER.

A personalised recovery plan.
The letter becomes each day's medicines, exercises, walks and appointments. Nobody types anything in.

It knows what is done.
"I'm back from my walk" ticks walk 1 of 2, and the plan updates.

It never guesses.
Fixed rules flag an unclear line, and AfterWard says "your letter does not say".

Reminds you at the right time.
Proactive voice reminders for medicines, appointments and daily tasks.

It shows its source.
The exact sentence from your discharge letter is on screen.

Why now: every hospital writes its letter differently.
A language model can read them all, and fixed rules make that safe to act on.

BUILD, SHIP, SHAPE: AMAZON DEVELOPER HACKATHON

One Letter Becomes a Recovery Plan

AfterWard turns a hospital discharge letter into a personalized, day-by-day plan for recovery.

1 Discharge letter

Upload Margaret's hospital discharge letter

SYNTHETIC DOCUMENT. NOT A REAL PATIENT. FOR DEMO AND TESTING ONLY.

Northbrook General Hospital
Trauma and Orthopaedics · Kingfisher Ward (Ward 4)
1 Example Road, Northbrook NB1 0AA (fictional)
Switchboard: 0000 000 000 · Ward: 0000 000 001 (fictional numbers)

DISCHARGE SUMMARY
Our ref: TC/SYN-0001/10-16
Date: Friday 16 October 2026

Patient:	Mrs Margaret Ellis	**Hospital no.:**	SYN-0001
Date of birth:	3 March 1958 (age 68)	**Address:**	12 Example Close, Northbrook NB2 0ZZ (fictional)
Admitted:	Monday 12 October 2026 (planned)	**Discharged:**	Friday 16 October 2026
Length of stay:	4 days	**Discharged to:**	Home, lives with her husband
Consultant:	Mr R. Sample, Consultant Orthopaedic Surgeon	**Allergies:**	**No known drug allergies**

DIAGNOSIS
Primary osteoarthritis of the right knee.

PROCEDURE
Right total knee replacement on Monday 12 October 2026. The operation went as planned. No complications.

MEDICINES TO TAKE AT HOME

Medicine	How to take it	Status
1. Paracetamol 500 mg tablets	Take 2 tablets four times a day (morning, midday, [illegible] evening, bedtime) for 14 days.	New
2. Codeine 30 mg tablets	Take 1 tablet up to four times a day, only if needed for pain. Stop after 5 days.	New
3. Apixaban 2.5 mg tablets	Take 1 tablet twice a day (morning and evening). Continue as directed. Your consultant will tell you [illegible] how long to continue [illegible].	New
4. Senna 7.5 mg tablets	Take 2 tablets at bedtime if constipated.	New

2 AI extraction & validation

AfterWard reads the letter, extracts key information and creates a structured plan

- **Medications** — Paracetamol, Codeine, Apixaban, Senna
- **Daily activities** — Exercises, walking, ice pack, rest
- **Appointments** — Physiotherapy, wound clip removal, consultant review
- **Restrictions** — No driving, no kneeling, no bath/swimming
- **Care instructions** — Wound care, diet, equipment

AI — Validated and structured into Margaret's personalized recovery plan with daily tasks and reminders.

3 Personalized recovery plan

A clear, day-by-day plan that Margaret can follow, with voice, screen and touch.

AfterWard | Patient: Margaret DEMO TIME Friday 23 October 7:00 am

SAT 17 – FRI 23
This week

6/7 Days completed
3/14 Tasks completed

CHOOSE A DAY

Sat 17	Sun 18	Mon 19	Tue 20	Wed 21	Thu 22	Fri 23
✓	✓	✓	✓	✓	✓	Today

✓ All done · ○ In progress · ○ Nothing recorded

A day is done when every task and medicine dose that day is recorded.

← Back to Today

Saved records, not a clinical exam

FRIDAY 23 OCTOBER · TODAY
Day details

- ○ Knee exercises from your physiotherapy sheet — Morning · 1 of 3
- ○ Knee exercises from your physiotherapy sheet — Afternoon · 2 of 3
- ○ Knee exercises from your physiotherapy sheet — Evening · 3 of 3
- ✓ Walking — 07:30 · 1 of 2 — Recorded complete 23 Oct 2026, 07:00
- ○ Walking — 15:00 · 2 of 2
- ○ Ice pack on the knee — Any time

Medications
4 medicines with dosing schedules

Exercises
3 times a day from physiotherapy sheet

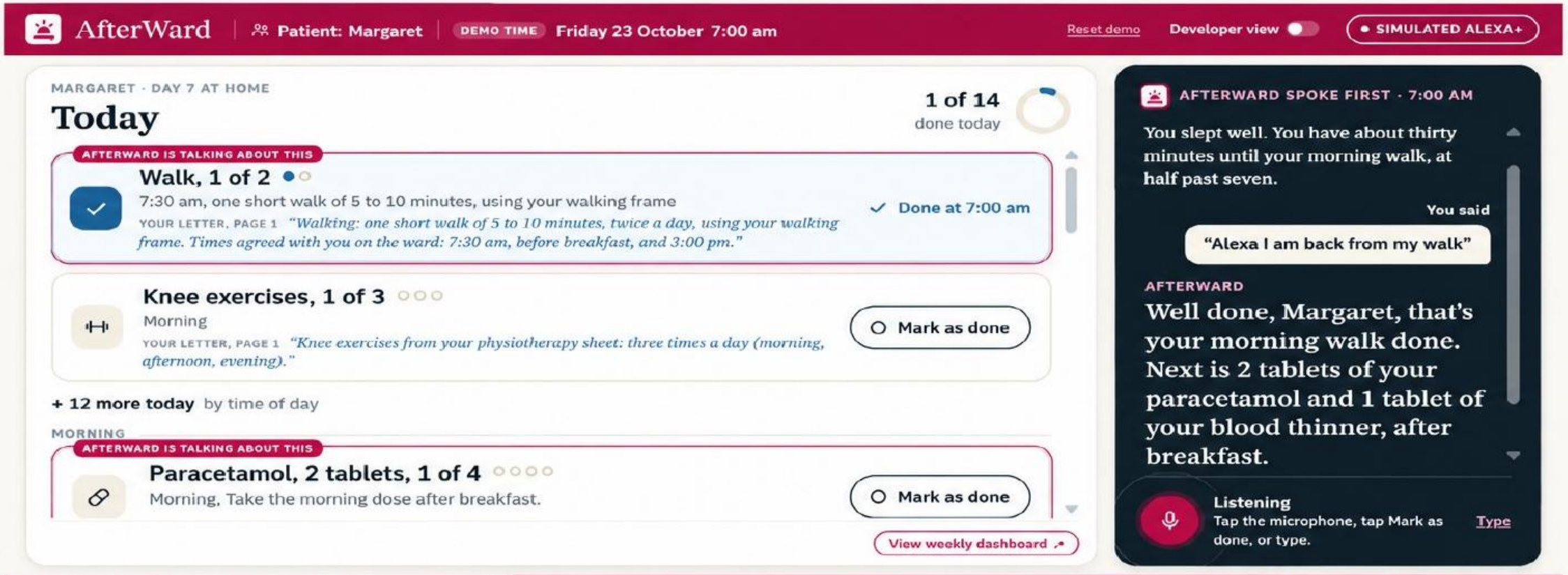
Appointments
Physiotherapy, wound clip removal, consultant review

Restrictions
No driving, no kneeling, no bath or swimming

Care instructions
Wound care, diet, equipment provided

ONE MORNING WITH MARGARET

A proactive reminder becomes a completed task—and the plan updates.



SPEAKS FIRST **RECORDS PROGRESS** **KNOWS WHAT IS NEXT**

AFTERWARD: A LIVING RECOVERY PLAN FOR ALEXA+

Beyond Reminders: A Plan That Remembers

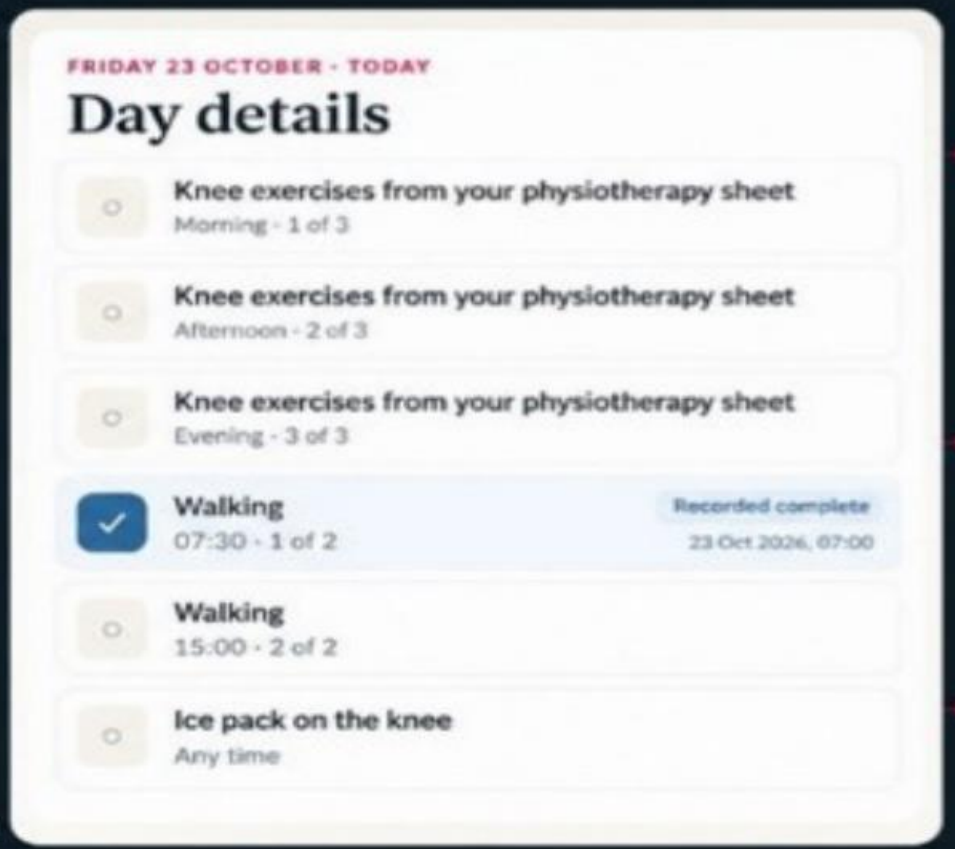
AfterWard turns your discharge instructions into a personalized, day-by-day plan.
It remembers what you've done, tracks your progress and helps you stay on track.

Weekly progress
See how many days and tasks you've completed.

Choose any day
View past and upcoming tasks at a glance.

Progress over time
AfterWard remembers your activity and builds a complete recovery history.





Personalized tasks
Your daily plan is built from your discharge letter.

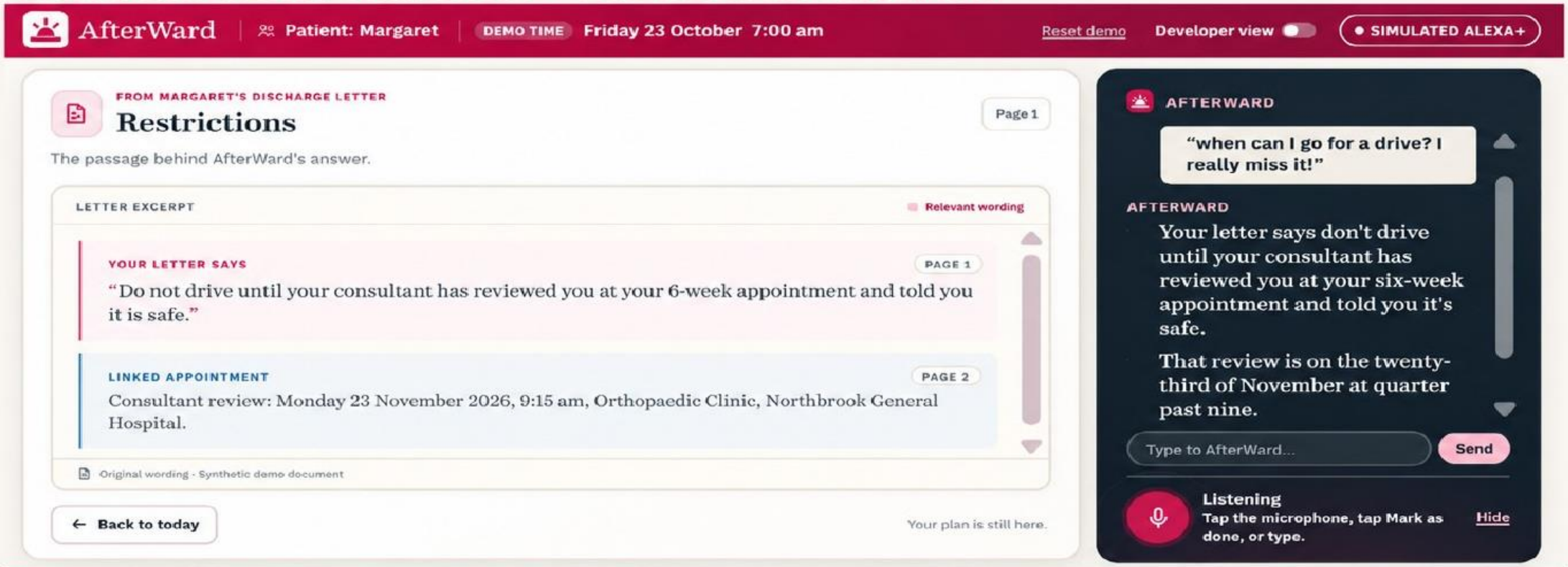
Track as you go
Mark tasks and medications as completed.

Nothing is missed
AfterWard keeps upcoming tasks and reminders on your plan.

Answers grounded in the discharge letter

Margaret asks a question. AfterWard shows exactly where the answer came from.

ACTUAL SIMULATED ALEXA+ SCREENSHOT · LETTER-GROUNDED ANSWER

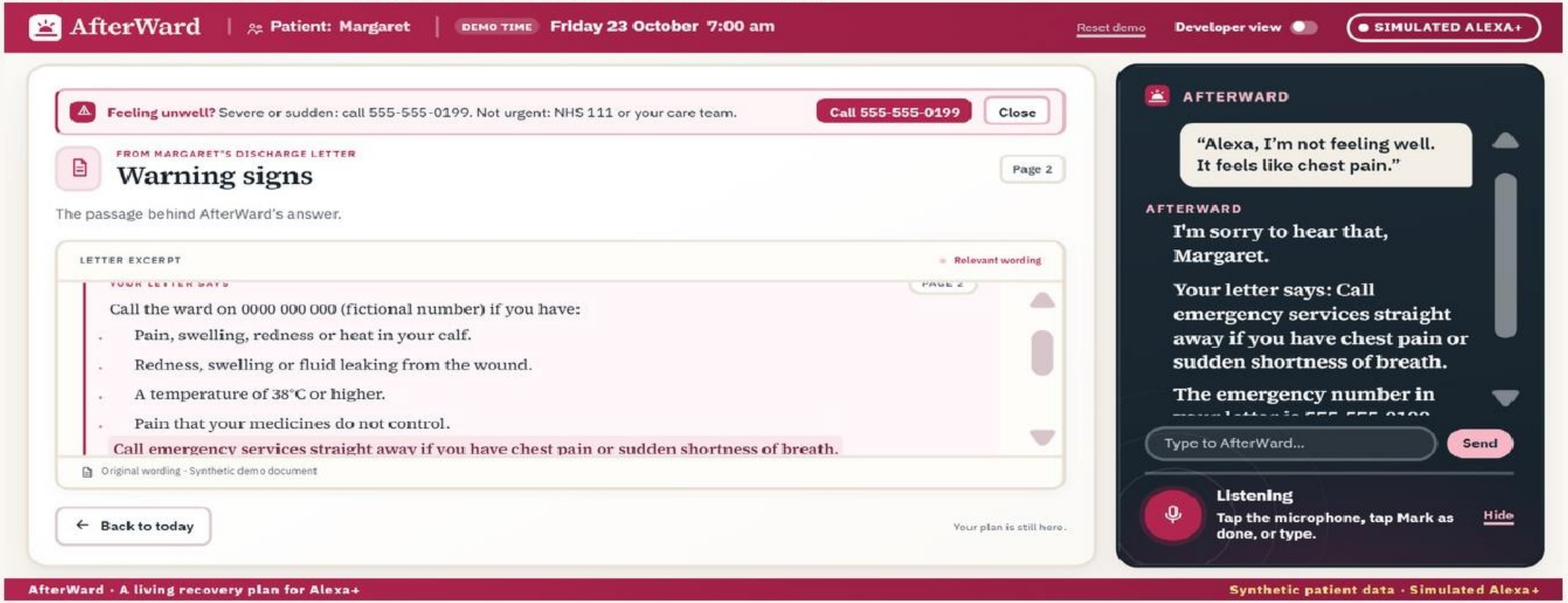


Driving guidance is linked to the restriction and consultant-review date in the letter.

Safety before automation

Urgent symptoms trigger safety guidance; the original letter remains visible.

ACTUAL SAFETY RESPONSE · WARNING SIGNS FROM THE LETTER



Fixed safety rules

Source-grounded answers

Uncertainty flagged

One plan. Three ways to interact.

Voice, screen and touch/text — one discharge-letter-grounded recovery plan.

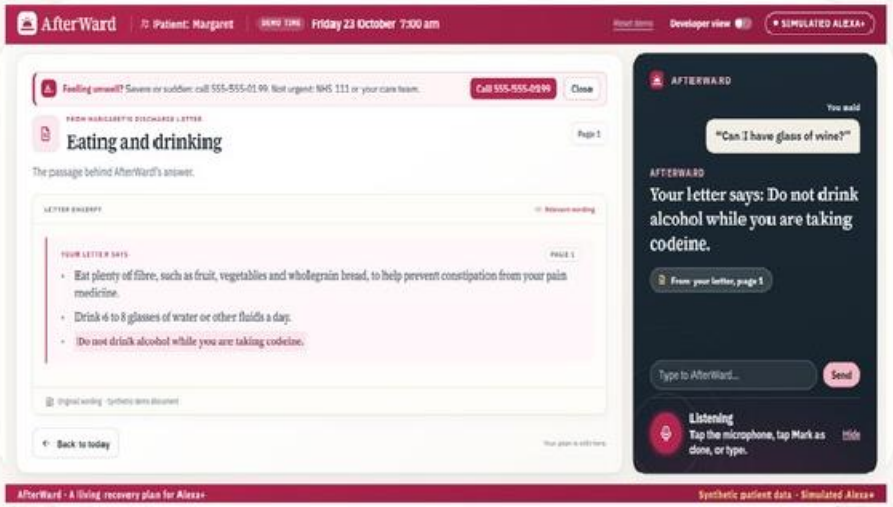
APPLICATION VIEW · VOICE, SCREEN, TOUCH + TEXT

AfterWard

● SIMULATED ALEXA+

One plan. Three ways to interact.

Margaret gets the same discharge-letter-grounded guidance through voice, screen or text.



01 VOICE

Hands-free questions and spoken answers — useful while Margaret uses a walking frame.

02 SCREEN

Readable recovery information and the relevant passage from her discharge letter.

03 TOUCH + TEXT

Type a question or tap an action. The same MCP-backed recovery plan powers each interaction.

AfterWard · A living recovery plan for Alexa+

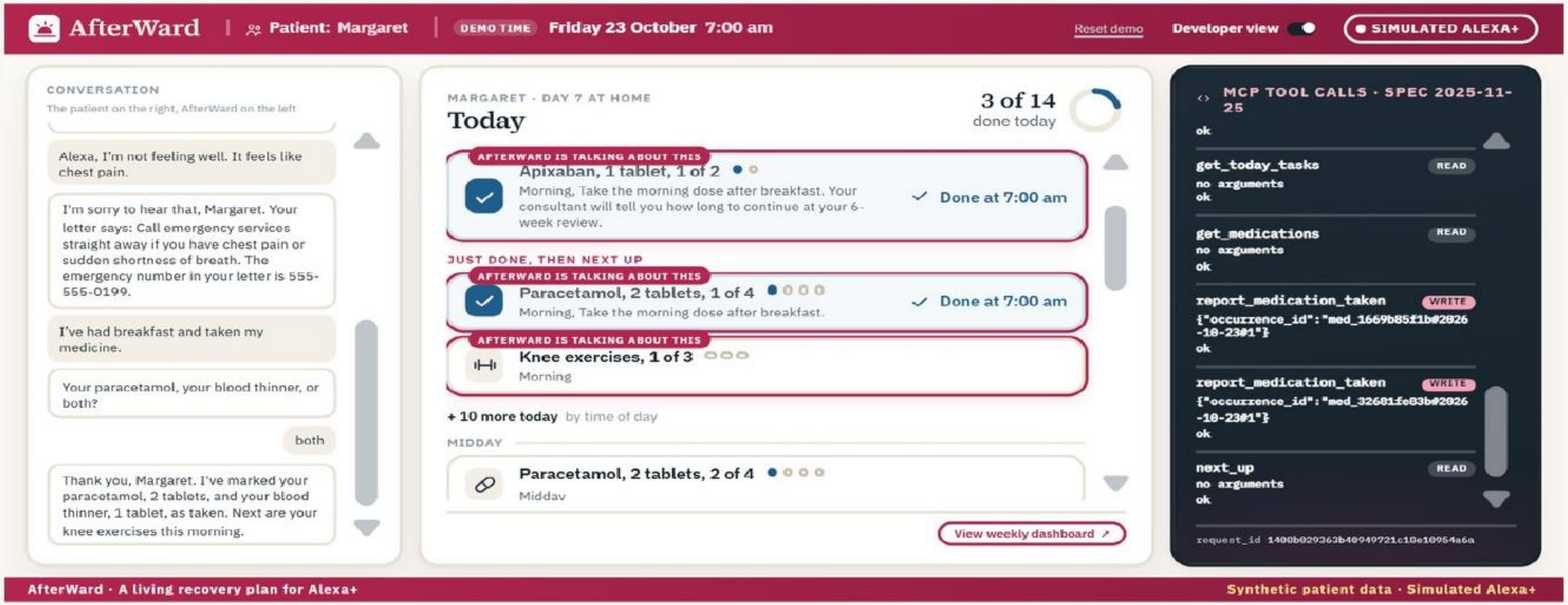
Synthetic patient data · Simulated Alexa+

Simulated Alexa+ client — not a live Alexa+ deployment.

MCP in action: ask → act → verify

A patient message calls MCP tools, updates saved state and refreshes Today.

REAL APPLICATION SCREENSHOT · CONVERSATION / TODAY / MCP CALLS



get_today_tasks · get_medications · report_medication_taken · get_recovery_plan

PROACTIVE MEANS MORE THAN "SET A REMINDER"

AfterWard speaks first

A reminder only knows the time. AfterWard knows her plan: what is done, what is next and what is still open.

1 Schedule

Amazon EventBridge checks every 5 minutes for anything due in the next 30 minutes.



Amazon EventBridge
Scheduler

Finds what's due
Looks at Margaret's plan and event history.

2 Guard

Checks if the task is already done, stopped or unclear. No reminder is sent.



Rule-based check
(Our safety rules)

Only send if needed
Uses her saved plan and latest updates to avoid unnecessary reminders.

3 Deliver

The client speaks a short, natural reminder. The words are built by code from her plan, not by the model.



Web/Alexa+ client
(SIMULATED)

Speaks proactively
For example: "It's time for your morning medicine."

Verified flow: EventBridge → Lambda → web companion speaks proactively (SIMULATED).
Classic Alexa skill does not speak first.

WHAT IS DONE, AND WHAT COMES NEXT

The MVP

A working end-to-end demo that turns a discharge letter into a safe, personalized recovery plan — with a clear path to what is next.

DONE

Built and running

A complete, working demo that shows the core value and safety approach.

- **Upload a discharge letter**
  It becomes a plan she can talk to.
- **Answers from her own letter**
  With the page they came from.
- **Safety gate (open source package)**
  Flags unclear items for a person to confirm.
- **Safe replies before any AI runs**
  A crisis check that replies safely.
- **Nine MCP tools, two prompts and an app**
  On spec 2025-11-25.
- **Proactive voice reminders**
  Reminder check runs on a schedule; the demo runs it locally.
- **Alexa skill and web voice client**
  With a text fallback.

SIMULATED

Simulated or synthetic

A realistic demo using simulated patients, data and Alexa+.

- **Alexa+ in a web app**
  The Alexa skill runs in Amazon's simulator.
- **Every discharge letter is synthetic**
  No real patient data.
- **One shared demo patient**
  Margaret, with a full recovery plan.
- **Fixed demo date**
  The demo clock is set to 23 October for the recording.

NEXT PHASE

What comes next

From a demo to a pilot and global rollout.

- **Secure sign-in and patient consent**
- **Reminder times she chooses**
  Even when her letter gives none (the letter stays in charge).
- **Unclear instructions to her care team**
- **Daily progress reports**
  For family and care team.
- **One-step calling to emergency services**
  Or her carer.
- **Photo letters, and voice in her own language**
- **Integration with hospital systems**
- **Clinical pilot before any real patient**

AfterWard is a proactive voice companion. Amazon EventBridge starts each reminder before it is due, so she never has to ask.

BUILT ON AWS · EU-WEST-2

How AfterWard works

From a discharge letter to a safe, personalized recovery plan — with proactive reminders and voice experience.

Green tags = hackathon requirement met

1 A letter becomes a plan

A discharge letter is turned into a safe, personalized plan.

Letter upload
Patient or hospital uploads discharge letter (about 15 seconds)

↓

Amazon S3
Private, encrypted bucket

↓

Amazon SQS
Queue, with a dead-letter queue and alarm

↓

AWS Lambda
Reads the letter's text, then Claude on Amazon Bedrock fills in a fixed form

↓

Safety gate (open source) ✓ OPEN SOURCE
discharge-letter-check (MIT)
Flags unclear items for a person to confirm

2 She talks to her plan

She asks about her plan using voice or screen, and AfterWard uses safe tools to help.

Simulated Alexa+ client ✓ SIMULATED ALEXA+
Next.js · voice in, voice out
plus an Alexa skill in Amazon's simulator

↓

Agent
Crisis check first: a fixed safe reply, no AI
Then Claude on Amazon Bedrock picks a tool

↓

AfterWard MCP server ✓ SELF-HOSTED MCP
Amazon ECS on Fargate, behind ALB + CloudFront
Streamable HTTP · spec 2025-11-25

↓

Nine tools, one patient
6 read: today, week, a day, medicines, plan, her letter
3 write: task done, medicine taken, undo
2 prompts · **MCP App**

3 AfterWard speaks first

Proactive reminders keep her on track, and she can always reply.

Amazon EventBridge Scheduler
Runs the reminder check every 5 minutes

↓

AWS Lambda
Finds what is due in the next 30 minutes

↓

Pending reminder
The client speaks it out loud, unasked.
Reminder text is built by code, not the model.

↓

She answers
"I'm back from my walk"
Ticks the walk off, saved in her event history

Data, kept per patient
Each patient's data stays separate and private, in the UK (eu-west-2).

Amazon DynamoDB
Recovery plan · event history

Amazon S3 Vectors
Her own letter · Titan Text Embeddings V2

✓ **AWS BUILDER**
14 AWS services, each documented

AfterWard is a proactive voice companion. Amazon EventBridge starts each reminder before it is due, so she never has to ask.

ALEXA+ TRACK · AWS BUILDER · OPEN SOURCE · EU-WEST-2

How AfterWard meets the brief

Every claim links to a file in the repo.

Green tags = hackathon requirement met

1 Patient & input channels

Margaret
talks or types

Simulated Alexa+
Next.js web app on screen

Mobile app PLANNED
Reminders, tasks and plan view

Hospital systems PLANNED
Discharge summary (PDF / HL7 / FHIR)

2 Edge & identity

Amazon CloudFront PLANNED
Security, DDoS, rate limits

Amazon Cognito PLANNED
User login, roles (patients, carers, clinicians)

Amazon API Gateway PLANNED
REST/HTTP endpoints for web, mobile, MCP

3 AfterWard MCP server (core services on ECS Fargate) R1

AfterWard MCP server
Python MCP SDK · 9 tools
ECS Fargate behind ALB + CloudFront

- get_today_tasks
- get_medications
- report_medication_taken
- get_recovery_plan
- get_week_summary
- search_discharge_instructions
- report_task_completed
- next_up
- get_patient_context

Safety gate (open source) R7
discharge-letter-check (MIT)
Validates source text, flags missing or unclear items

4 Data layer

Amazon S3 Vectors R3
Her letter, filtered by patient (Titan Text Embeddings V2)

DynamoDB: event history R3
Append only log of what she did

DynamoDB: recovery state R5
Plan, today's tasks, next up and extracted documents

Amazon S3 R5
Discharge letters (PDF) and extracted documents

Amazon SQS R4
Task and reminder queue (fan-out to workers)

Amazon EventBridge R4
Schedules, daily summaries, follow-ups

5 AI / ML services

Amazon Bedrock R6
LLM for grounded answers (using RAG + letter context)

Reminder Lambda R4
Every 5 minutes, saves a check-in for AfterWard to speak

Bedrock extraction + discharge-letter-check R5
Extract text and structure from discharge letters

Hackathon requirements

Status today, with the file that proves it

R1	**Self-hosted MCP server, spec 2025-11-25** Streamable HTTP, on ECS Fargate app/mcp/server.py	MET
R2	**MCP server really called in code** Every read and write goes through it app/mcp/client.py	MET
R3	**Simulated Alexa+ experience, with source** Labelled web app + Alexa skill in app/skills/	MET
R4	**Public repo with an open-source licence** MIT licence in LICENSE + repository is public	MET
R5	**Clear setup and run instructions** README for judges, checked by a fresh-clone test	MET
R6	**AWS Builder: services with documented use** 14 AWS services, each linked to its file - docs/aws-integrations.md	MET
R7	**Open source: a new, separate project** discharge-letter-check · MIT · v0.1.0 149 tests - used by AfterWard	MET

Tested: +1,200 Python · 100+ web

Tech choices and the trade-offs we accepted

Two models, one switch
Sonnet 4.5 (final runs)
Haiku 4.5 (testing)
Best quality for final runs; faster for testing.

Rules gate, not an AI checker
Open-source Python package
Deterministic, testable and safe.

S3 Vectors, not OpenSearch
Simple and reliable for our scale.
Swappable if needed.

DynamoDB, not SQL
Fast patient state and task history.
Designed for high concurrency.

MCP on Fargate, not Lambda
Multiple tools, state and scheduled jobs.
Easier to maintain.

Browser voice, not Polly
Alexa+ voice when access opens.
Polly is not used for testing or demos.

THE BUILD / ARCHITECTURE → OPEN SOURCE

One recovery system. Two public repositories.

Architecture components map directly to accessible, independently reviewable source code.

01 AFTERWARD APPLICATION

Alexa+ simulated client + MCP server

Main application

Alexa+ simulated client
Next.js web app with voice interface

MCP server & tools
9 MCP tools · 2 prompts · MCP App

AWS integration
S3 ingestion, DynamoDB, EventBridge, Lambda

Reminders & recovery state
Tracks tasks, medications, walks and appointments

OPEN PUBLIC REPOSITORY
post-discharge-mcp

02 DISCHARGE LETTER CHECK

Reusable structured-letter validation

Validation package

Structured letter validation
Parses discharge letters (PDF / HL7 / FHIR)

Flags missing, unclear or inconsistent fields
Validates key sections using deterministic rules

Reusable package
MIT-licensed, usable independently

Integrates with AfterWard
Used in ingestion pipeline for safety and reliability

OPEN MIT-LICENSED PACKAGE
discharge-letter-check

ARCHITECTURE FLOW
From letter to patient experience

Discharge letter (PDF / HL7 / FHIR) → S3 ingestion (documents) → Lambda (processing) → Validation package (discharge-letter-check) → DynamoDB plan (recovery state) → MCP tools (patient experience)

IMPACT

North Star: Recovery Days on Track

Record what happened — do not confuse it with a clinical outcome.

PATIENT VALUE

Everyday confidence
See the tasks and medicines due today.

Receive planned reminders
without asking.

Ask questions in her own words
grounded in her letter.

NORTH STAR

Days on track
A day counts when all scheduled items are recorded done.

Missing records mean unknown, not missed.

Weekly history is activity, not a clinical score.

NEXT EVIDENCE

Measure in a pilot
Letter-to-plan time and reminder responses.

Source-grounding and safety guardrails

Clinician review
A clinical pilot before outcome claims.

No claim of fewer readmissions: that requires evaluation with real patients.
AfterWard measures task completion and engagement to support everyday recovery.

IMPACT AND GROWTH

Roadmap and global expansion

From hackathon MVP to real-world impact, with a path to global reach.

Now

Hackathon MVP

LETTER TO DAILY PLAN

- Open-source safety checks
- 9 MCP tools, 2 prompts, an MCP App
- Scheduled reminders
- Simulated Alexa+ and a classic Alexa skill

Next

Ready for a pilot

SECURE SIGN-IN AND CONSENT

- Secure sign-in and consent
- Reminder times the patient can change
- A family view, with permission
- Rules reviewed by a clinician
- An Alexa+ add-on, when the toolkit opens

Later

Connected

HOSPITAL LINK (FHIR)

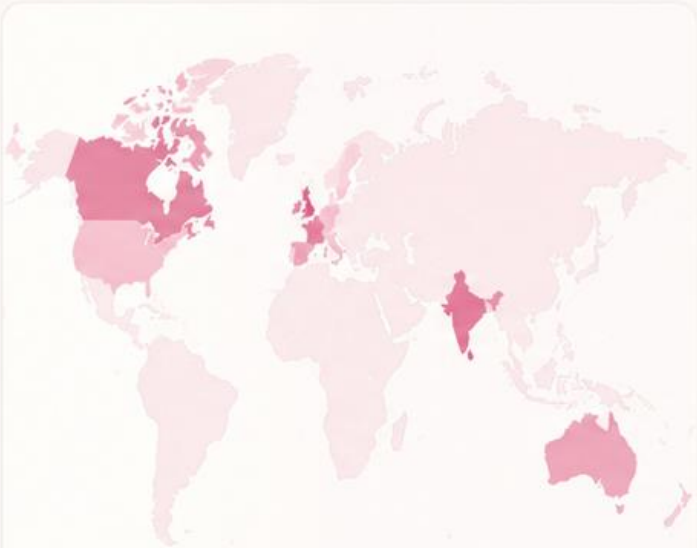
- Hospital link (FHIR), so the letter arrives digitally
- A care team view, with permission

Global

Country by country

LOCAL AND SAFE

- One AWS region per country, so data stays local
- Local languages and medicine names
- Local privacy rules: UK GDPR, HIPAA, GDPR
- A medical device check in each market



Global expansion

Built to work across regions, languages and healthcare systems.

Everything after "Now" is a plan.

The safety checks and the MCP tools carry over to each country. The region, the language and the rules are what change.

TECHNOLOGY CHOICES – WHAT WE BUILT, WHY, AND THE TRADE-OFFS

Key technology choices across After Ward

We selected tools and services that balance safety, cost and scalability, with clear trade-offs.

Models, data and processing

Decision	We chose	Why	Trade-off we accept
Reading the letter	Claude Sonnet 4.5 on Bedrock (forced tool output)	Reads messy letters well; predictable JSON; nothing to host.	Cost per letter output can vary, so the safety gate is required.
Agent (demo only)	Claude Sonnet 4.5 on Bedrock	Best at choosing tools and asking "which tablet?"	Slower and pricier than Haiku.
Dev and test model	Claude Haiku 4.5 (EU inference profile)	Cheap and fast for many test runs; stays in EU regions.	Lower quality, so final tests rerun on Sonnet.
Model switching	One MODEL_MODE setting	Swaps every model with no code change.	Two sets of models to keep tested.
Vector store (embeddings)	Titan Text Embeddings V2, 512 dimensions	AWS-native, cheap, fits S3 Vectors.	Less precise than, 1,024 dimensions; the size is fixed for life.
Database	DynamoDB: Recovery State and Event History	Fast reads by patient; conditional writes; pay per request.	No joins or flexible queries.

Platform, experience and safety

Decision	We chose	Why	Trade-off we accept
AWS region	eu-west-2 (London)	Close to UK users; patient data stays in the UK.	Newer services and models reach London later.
MCP server	Python MCP SDK, Streamable HTTP (spec 2025-11-25)	Required by the track; testable with MCP Inspector.	A public endpoint we must secure (token and Origin check).
Hosting	A container on ECS Fargate behind a load balancer and CloudFront	The AgentCore starter would not deploy; a container runs the same anywhere.	An always-on task to pay for and keep patched.
Infrastructure	Task definition, Lambda build script and alarm template in the repo	Every integration visible in code, and listed in one AWS page.	Not one-click yet; some steps run by hand.
Monitoring	CloudWatch: JSON logs, 5 metrics, 3 alarms with email	Built in; a failed letter or tool error reaches us at once.	AWS-only; no dashboard yet.
Patient interface	Next.js web app: simulated Alexa+, voice and screen	Shows the whole experience; the Alexa+ toolkit is in preview.	Must always be labelled as simulated; no production use yet.
Voice in and out	Browser's speech (Web Speech API)	Her hands are busy, so she talks and replies are spoken.	Sounds different on each computer.
Safety checks	discharge-letter-check (open source library)	Fixed rules, 149 tests, flag unclear items for a person to confirm.	Only catches problems we have written rules for.

HOW IT IS BUILT: A DESIGN SKETCH, NOT BUILT

Architecture at scale

The same flow works for one patient or millions. We scale each stage, but keep the design simple and modular.

A — Talking to the plan
Many patients at once.

Patients (many at once) → Alexa+ or any MCP client (the agent) → Load balancer (spreads the calls) → MCP server copies (identical, no shared state)

B — A letter becomes a plan
Uploads handled in seconds, even at peak.

Letter uploads (about 10 seconds at peak) → Amazon S3 (private bucket) → Amazon SQS (absorbs the bursts) → Lambda workers (about 200 at once)

C — AfterWard speaks first
The same reminder logic, spread over many workers.

Schedules (per patient and time zone) → Amazon EventBridge (a schedule per time zone) → Queue (batches of patients) → Reminder workers (many at once)

Data, kept per patient

- **Recovery State** — DynamoDB: one small item per medicine, task and daily occurrence
- **Event History** — DynamoDB: what each patient did
- **Amazon S3 Vectors** — search, filtered to one patient

Pink boxes are what changes at scale. The shape of the system stays the same. Each service runs independently, so any one choice can be swapped for its open-source option by replacing one module.

WHAT'S NEXT · TARGET ARCHITECTURE · DATA STAYS IN THE EU

AfterWard at scale: built for millions of patients

Same MCP tools and safety gate as today. Designed for scale, security and reusability.

EU-WEST-2
Data stays in the EU

Open source
MIT licensed

Scalable
Millions of patients

1 Patient & input channels

Margaret
talks or types

Simulated Alexa+
Next.js web app on screen

Mobile app PLANNED
Reminders, tasks and plan view

Hospital systems PLANNED
Discharge summary (PDF / HL7 / FHIR)

2 Edge & identity

Amazon CloudFront + PLANNED
Security, rate limits

Amazon Cognito PLANNED
User login, roles (patients, carers, clinicians)

Amazon API Gateway PLANNED
REST/HTTP endpoints for web, mobile, MCP

3 AfterWard MCP server (core services on ECS Fargate) R1

AfterWard MCP server
Python MCP SDK · 9 tools
ECS Fargate behind ALB + CloudFront

- get_today_tasks
- get_medications
- report_medication_taken
- get_recovery_plan
- get_week_summary
- search_discharge_instructions
- report_task_completed
- next_up
- get_patient_context

Safety gate (open source) R7
discharge-letter-check (MIT)
Validates source text, flags missing or unclear items

4 Data layer

Amazon S3 Vectors R3
Her letter, filtered by patient (Titan Text Embeddings V2)

DynamoDB: event history R3
Append only log of what she did

DynamoDB: recovery state R5
Plan, today's tasks, next up and extracted documents

Amazon S3 R5
Discharge letters (PDF) and extracted documents

Amazon SQS R4
Task and reminder queue (fan-out to workers)

Amazon EventBridge R4
Schedules, daily summaries, follow-ups

5 AI / ML services

Amazon Bedrock R6
LLM for grounded answers (using RAG + letter context)

Knowledge base PLANNED
Guidelines, discharge instructions and policies

Reminder Lambda R4
Every 5 minutes, saves a check-in for AfterWard to speak

Bedrock extraction + discharge-letter-check
Extract text and structure from discharge letters

Operations & security

AWS IAM, encryption at rest
Encryption at rest, least privilege access

Amazon CloudWatch
Logs, metrics, alarms, cost monitoring

CI/CD (GitHub Actions) PLANNED
Tests, build, deploy (ECS Fargate, Lambda)

Observability PLANNED
Tracing, error tracking, usage analytics

Data governance
EU data residency, access controls, audit logs

Tech choices and the trade-offs we accepted

Two models, one switch
Sonnet 4.5 (final runs)
Haiku 4.5 (testing)
Best quality for final runs; faster for testing.

Rules gate, not an AI checker
Open-source Python package
Deterministic, testable and safe.

S3 Vectors, not OpenSearch
Simple and reliable for our scale.
Swappable if needed.

DynamoDB, not SQL
Fast patient state and task history.
Designed for high concurrency.

MCP on Fargate, not Lambda
Multiple tools, state and scheduled jobs.
Easier to maintain.

Browser voice, not Polly
Alexa+ voice when access opens.
Polly is not used for testing or demos.

Hackathon requirements — status

R1	Self-hosted MCP server, spec 2025-11-25	MET
R2	9 MCP tools implemented	MET
R3	Simulated Alexa+ client (web)	MET
R4	Public repo with open-source licence (MIT)	MET
R5	Clear setup and run instructions (README)	MET
R6	AWS Builder with documented services	MET
R7	Separate validation package (discharge-letter-check)	MET

Tested: 1,269 Python · 138 web

HOW IT IS BUILT: IF EVERY US HOSPITAL ADMISSION USED AFTERWARD

How it scales

Designed to handle national hospital volume, with the same simple architecture, just more of each service.

10

uploads a second at peak

Discharge letters from hospitals across the US.

5,000

tool calls a second at peak

Plan lookups, task updates, reminders and other tools.

Assumptions, not measured

- 35.7 million admissions a year (AHA Fast Facts, 2024 survey)
- 30 days of use per patient
- 15 tool calls a day
- Peak is 10× the average
- 20 seconds to read a letter

Part	Hackathon build (today)	At national scale (target)
Letter workers	2 at once	About 200 at once, capped to the Bedrock limit
Amazon Bedrock	Default request cap	Cap raised to about 600 letters a minute
MCP server	One deployment	Several identical copies behind a load balancer
Plan storage	One item per patient	One small item per medicine, task and daily occurrence
Reminders	One schedule, one Lambda	A schedule per time zone and a queue of patient batches

A design sketch, not built. The architecture stays the same — it scales by running more of each service in parallel.

THE JOURNEY: REAL LESSONS FROM BUILDING

We didn't just build it. We tested the boundaries.

15 documented friction points · hands-on AWS and Alexa+ feedback · actionable fixes

FRICTION LOG

Alexa+ access: a working MCP, no partner sandbox

Expected	Connect self-hosted MCP server to Alexa+.
Reality	MCP Toolkit limited to selected partners.
Workaround	Clearly labelled simulated Alexa+ browser client plus tested classic Alexa skill using same MCP tools.
Ask to Amazon	Open an MCP Toolkit sandbox for hackathon teams.

PRODUCT FEEDBACK

S3 Vectors: correct retrieval, silent empty results

Worked	Serverless per-patient letter retrieval with page citations.
Friction	Old index configuration returned zero results without warning (~2 hours).
Fix	Align vector bucket/index configuration.
Ask to AWS	Warn when querying an index without matching vectors.

WHAT BUILDING TAUGHT US

Bedrock: require retrieval before "not in letter".

Extraction: surface unmapped safety-critical lines.

Voice: short utterances need alternatives + confidence.

Engineering feedback from synthetic demos—not patient research · Full logs and product feedback in public docs.

THE JOURNEY: A FIRST-TIME TEAM ON AN AGRESSIVE SCHEDULE

Challenges we faced

Real challenges, real learning — and how we adapted to keep moving forward.

1 First time on AWS

This was our first build on AWS. We had to learn each service, from Bedrock to Lambda and DynamoDB, while building the application.

→ **Gained hands-on experience and built step by step.**

2 A student developer

Our developer is a final-year Computer Science student. AWS and voice agents were new to us, so we learned them quickly while building.

→ **Turned a steep learning curve into valuable progress.**

3 Monitoring at full speed

Guiding a student through a first cloud build meant teaching and delivering at the same time.

→ **Set up simple, effective monitoring and alerts.**

4 An aggressive schedule

We learned the stack and shipped it in the same weeks. A fixed cut-off date meant safety and simplicity were non-negotiable.

→ **Focused on core features and kept the design simple.**

5 A safety rule too blunt

It would have hidden a blood thinner over an unclear end date. We now flag the field, not the whole item.

→ **Refined the rule to be specific and less disruptive.**

6 Queues deliver twice

A repeated message must change nothing. Every write carries a key, so a repeat is saved once.

→ **Made all steps idempotent and safe to repeat.**

7 No Alexa+ to test on

The Alexa+ toolkit is in preview with select partners. We built a web client and always labelled it as simulated.

→ **Used Amazon's simulator and kept it clearly labelled.**

Each challenge made AfterWard stronger. We kept safety, simplicity and patient focus at every step.

THE JOURNEY: OUR FIRST BUILD ON AWS, AND OUR FIRST VOICE AGENT

What we learned

Key lessons from our first build — on AWS and with voice — that shaped a safer, stronger product.

On AWS

FROM SETUP TO SCALE

Check the region on day one

Newer services and models reach London later. One read-only call per service proves the setup.

Some choices cannot be undone

A vector index keeps its size for life, so we fixed 512 dimensions before indexing anything.

Assume every message arrives twice

A queue can redeliver, so each write has to be safe to repeat.

Test cheap, finish strong

A small model for daily tests and the final models for the last run, swapped by one setting.

On voice

FROM WORDS TO ACTION

Ears are not eyes

A long spoken list is hard to follow. AfterWard names the next two items and shows the rest on screen.

Ask before you write

“I took my tablets” could mean any of them. AfterWard asks which one and saves nothing until it knows.

A reminder is not a question

“I hope you slept well” is a statement on purpose. AfterWard must not judge symptoms.

The rules live in the tools

With real Alexa+, Alexa+ is the agent. So every rule sits in the MCP tools, not in our demo agent.

These lessons turned early roadblocks into a more reliable, safer and more realistic product.

Real-world constraints — cloud setup, data limits, ambiguous speech and safety rules — shaped AfterWard from the start.

THE JOURNEY: HOW AFTERWARD MEETS EACH ONE

Judging criteria

A complete, working prototype that is safe, useful and built with the required tools.

TECH IMPLEMENTATION
How well is the project built?

- Self-hosted MCP server on ECS Fargate (eu-west-2)
- 14 AWS services integrated with IAM roles and encryption at rest
- 9 tools, 2 prompts and an MCP app (MCP Inspector & MCPJam)
- 1,200+ Python and 100+ web tests pass

DESIGN
Does it deliver a complete, coherent product experience?

- Voice first, because her hands are on a walking frame
- A simple, clear screen that mirrors her voice experience
- One clear flow from daily reminders to a living recovery plan
- Also works by text for flexibility and accessibility

POTENTIAL IMPACT
Does it solve a real, specific customer need?

- One named user and one specific need (post-discharge recovery)
- Every number has a source or is our own test
- A North Star metric: recovery days on track (to measure in a pilot)
- Helps older adults recover safely at home with confidence and independence

QUALITY OF THE IDEA
Is this a creative, imaginative use of the required tools?

- More than a reminder or document Q&A
- A plan that knows what is done and flags what it cannot confirm
- The safety gate is a reusable open-source package
- Combines Alexa+, MCP, AWS and real-world safety in a new way

Together, these criteria show a complete, safe and impactful solution.

A real use case, working implementation, thoughtful design and measurable impact — built with the required tools.

THE JOURNEY: HOW AFTERWARD MEETS EACH ONE

Mini challenges

Two important areas we built from scratch during the hackathon, and what we learned.

AWS BUILDER

Goal	Build a secure, scalable backend for a real post-discharge use case, using AWS services.
What we did	Integrated 14 AWS services in eu-west-2 (London). Documented each service, its code and live status.
Key learnings	Check the region on day one. Newer services and models reach London later. Plan for permissions, logging and monitoring early.
Result	A working, documented cloud setup with monitoring and alarm rules.

AWS SERVICES USED (14)

Bedrock · S3 · S3 Vectors · SQS · Lambda · DynamoDB · EventBridge · ECS Fargate · ECR · Load Balancing · CloudFront · Secrets Manager · CloudWatch · Titan Embeddings

OPEN SOURCE

Goal	Create a reusable safety check for discharge letters that others can use.
What we built	A new public repository with MIT license, version 0.1.0, with a fixed form for each letter and rules to flag missing, unclear or unreadable lines.
Validation	149 tests passing on Python 3.11 to 3.13, run by GitHub Actions.
Impact	AfterWard uses this package, so the open-source code is the real safety check, not just a demo.

REPOSITORY

`github.com/charvimaurya/discharge-letter-check`

MIT license · v0.1.0 · 149 tests passing · Python 3.11 – 3.13 · CI with GitHub Actions

Together, these mini challenges taught us to plan early, validate continuously, and keep safety at the core.

Real-world constraints — AWS regions, permissions, noisy letters and voice inputs — shaped a more robust and reliable product.

Public, open source, and testable

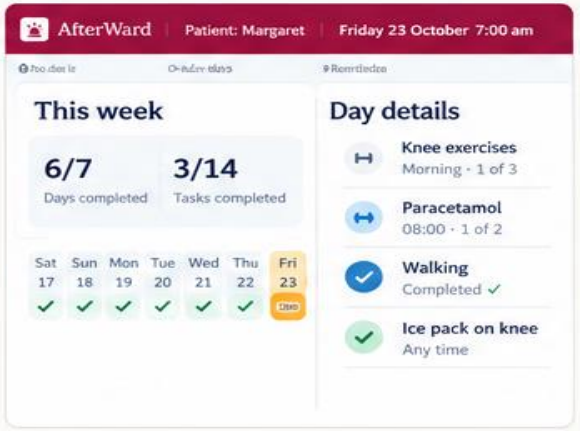
Two repositories: the main Alexa+/MCP application and a reusable validation package.

AfterWard — main application

github.com/charvimaurya/
post-discharge-mcp

MCP server for a living recovery plan.

- 9 MCP tools and 2 prompts
- AWS integrations (S3, DynamoDB, ECS)
- MIT licence
- README & local setup guide



discharge-letter-check

github.com/charvimaurya/
discharge-letter-check

Reusable discharge-letter validation package.

- 149 tests (Python 3.11–3.13)
- Validates discharge instructions
- MIT licence
- Published as open-source package



ARCHITECTURE SLIDE: REQUIREMENTS R1–R7

R1 Self-hosted MCP server	R2 MCP server really called in code	R3 Simulated Alexa+ experience	R4 Public repo	R5 Clear setup and run steps	R6 AWS Builder: services documented	R7 Open source: separate project
MET	MET	MET	MET	MET	MET	MET

AFTERWARD: A LIVING RECOVERY PLAN FOR ALEXA+

Thank you

to the Alexa team and Amazon Developer.

For a wonderful opportunity to learn and build a unique experience at Build, Ship, Shape.
We'd also love the chance to build what comes next because, after all,
no one should have to remember their recovery alone.



Real people
Inspired by patients like Margaret, and everyone recovering at home.

Responsible AI
Grounded in their own medical information, with safety built in.

Bigger impact
A step towards more supported recoveries, aims to support fewer readmissions and healthier lives at home.

AfterWard (main app)
github.com/charvimaurya/post-discharge-mcp

Open source: discharge-letter-check
github.com/charvimaurya/discharge-letter-check

Team: Shilpa Agarwal and Charvi Maurya

A living plan: reminds, records progress, and flags uncertainty for a person.

Discharge is one moment. Recovery happens every day.